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Gesture recognition from surface electromyography signals based on the SE-DenseNet network

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Abstract

Objectives: In recent years, significant progress has been made in the research of gesture recognition using surface electromyography (sEMG) signals based on machine learning and deep learning techniques. The main motivation for sEMG gesture recognition research is to provide more natural, convenient, and personalized human-computer interaction, which makes research in this field have considerable application prospects in rehabilitation technology. However, the existing gesture recognition algorithms still need to be further improved in terms of global feature capture, model computational complexity, and generalizability.
Methods: This paper proposes a fusion model of Squeeze-and-Excitation Networks (SE) and DenseNet, inserting attention mechanism between DenseBlock and Transition to focus on the most important information, improving feature representation ability, and effectively solving the problem of gradient vanishing.
Results: This proposed method was tested on the electromyographic gesture datasets NinaPro DB2 and DB4, achieving accuracies of 85.93 and 82.39 % respectively. Through ablation experiments, it was found that the method based on DenseNet-101 as the backbone model produced the best results.
Conclusions: Compared with existing models, this proposed method has better robustness and generalizability in gesture recognition, providing new ideas for the development of sEMG signal gesture recognition applications in the future.

Keywords: gesture recognition; EMG signal; DenseNet; attention mechanism; deep learning

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Introduction

Electromyographic signals are a type of bioelectrical signal generated during muscle contraction movements. These signals can be generated within muscles or on the skin surface and are acquired by invasive and non-invasive methods, namely electromyography (EMG) and surface electromyography (sEMG). The EMG obtained through invasive methods is collected by needle electrodes, including monopolar single electrode, single fiber EMG electrode, and concentric EMG electrode [1]. The sEMG obtained by non-invasive methods is a technique that records muscle electrical activity by placing surface electrodes on the skin surface. There are two types of surface electrodes: gel EMG electrode and dry EMG electrode [2]. During muscle contraction or relaxation, neurons within the muscle generate electrical signals and produce small potential changes on the skin surface, which are captured by electrodes and converted into digital signals for analysis and processing. Unlike invasive EMG signals, sEMG signals do not require insertion into muscle tissue, making signal acquisition more convenient and comfortable, and facilitating real-time monitoring.

With the advancement of science and technology, EMG signals have become increasingly important in human–computer interaction. Their application areas mainly include gesture recognition [3], muscle control [4], biofeedback [5], emotion recognition [6, 7], and gaming entertainment. In recent years, research on gesture recognition based on sEMG signals has received widespread attention and has been applied in multiple fields such as prosthetic control [8] and rehabilitation robots [9]. The sEMG signals can directly reflect muscle activity and capture subtle changes in muscle relaxation and contraction. This recognition ability is mainly reflected through the specificity of muscle groups and the uniqueness of signal features. Different gestures usually activate different muscle groups, and each sEMG signal contains different frequency and amplitude changes, resulting in different characteristics of sEMG signals.

In preliminary studies, scholars identified gestures by extracting surface electromyographic signal features and employing machine learning for classification. The features extracted by machine learning methods include time-domain features (such as mean and variance), frequency-domain features (such as power spectral density), and time-frequency domain features (such as wavelet packet coefficients). The commonly used classifiers include Support Vector Machines (SVM), K-Nearest Neighbors (KNN), and Decision Trees.

In 2010, Arjunan SP proposed a novel method for machine learning based on fractal features of sEMG signal – fractal dimensions and maximum fractal length, using a dual support vector machine (TSVM) to decompose the classification task into two SVMs for each class [10]. The Artificial neural network (ANN) models perform better in gesture recognition classification tasks, achieving relatively high classification accuracy [11, 12]. Nagarajan et al. [13] extracted a multi-class method based on Partial Least Squares Discriminant Analysis (PLS-DA), which significantly improved hand motion accuracy compared to linear discriminant analysis methods. Gozzi et al. [14] proposed applying explainable AI (XAI) to gesture recognition classification tasks for feature selection, with three features being more effective.

With the rapid development of deep learning in recent years, many researchers have introduced it into the study of sEMG gesture recognition [15–19]. Deep learning can better process large amounts of data and automatically learn feature representations, thus avoiding the tedious process of manual feature extraction. This makes deep learning more effective in sEMG signal gesture recognition. Atzori et al. [20] used convolutional neural networks to classify and recognize sEMG signals, demonstrating that a simple convolutional neural network architecture can produce results comparable to classical classification methods. Geng et al. [21] introduced the concept of sEMG images composed of high-density sEMG (HD-sEMG) space and used deep convolutional networks for classification. Without any window features, the single frame recognition accuracy of eight8 gestures reached 89.3 %. Also in a study based on HD sEMG, Yu et al. [22] designed a domain adaptation framework based on deep learning to improve inter-session gesture recognition, achieving an accuracy of 85 % in cross-session and cross-topic scenarios. In the studies based on sparse multi-channel sEMG, Wei et al. [23] observed that a few muscles play an important role in specific gestures, learned their relevance through a "divide-and-conquer" strategy, and proposed a multi-stream convolutional neural network (CNN) framework. Ding et al. [24] proposed a parallel multi-scale convolution architecture that utilizes larger kernel filters, and the final results showed that it can significantly improve classification accuracy. Tsinganos P treated the sEMG signal gesture recognition problem as a sequence classification problem and used a temporal convolutional network, which improved the results by nearly 5 % [25]. Kumar NK [26] and Hu [27] both considered the temporal characteristics of sEMG signals. The former proposed a hybrid architecture based on CNN and Bidirectional LSTM (Bi-LSTM) to learn the channel and bidirectional temporal information of sEMG signals, while the latter used an attention-based hybrid CNN and RNN (CNN-–RNN) architecture to extract hidden correlations between different channels. Moslhi et al. [28] proposed that the Singer Transformer can better integrate the global information of signals and combine it with attention mechanisms to improve classification accuracy. Lee et al. [29] designed a classification model based on Graph Convolutional Network (GCN), which can effectively extract and aggregate key features related to various gestures. Islam et al. [30] proposed a lightweight All-ConvNet + transfer Learning (TL) model to enhance gesture recognition performance between conversations and subjects.

Traditional CNNs typically rely on layer-by-layer feature extraction, which may result in the loss of some shallow feature information. Unlike traditional convolutional networks, ResNet uses skip connections to alleviate gradient vanishing, but there is still room for improvement in feature reuse. This would result in a lower recognition rate, and its generalizability still needs to be improved. To address this issue, this study proposes an sEMG signal gesture recognition model based on SE-DenseNet. DenseNet uses dense network connections to transmit and reuse features, which helps preserve shallow features and achieve excellent parameter utilization. This reduces redundant information, thereby improving the learning and generalizability of the model, effectively alleviating the gradient vanishing problem and also reducing storage requirements. Dense connections are achieved through feature fusion at various levels, with each layer directly obtaining feature representation from the previous layer. This helps to learn of richer feature representations and the effectively capture multi-scale information, ensuring comprehensive information transmission and sharing. Inserting attention mechanisms between dense blocks and transitions could significantly enhances the feature representation ability of the model, which is crucial for handling complex tasks such as image classification.

The rest of this article is roughly as follows: "Materials and data preprocessing" section introduces the process of data processing; "Methods" section provides a detailed introduction to the SE-DenseNet classification model; "Results and discussion" section presents the experimental results of the algorithm; "Conclusions" section summarizes the research of this article and proposes further work.

Materials and data preprocessing

Dataset

The experimental data used in this article comes from the Ninapro database [31, 32], which is also a commonly used dataset by researchers in this field. In order to fairly compare existing models, this study conducted experiments using the same data subset DB2 and DB4 as in the references [33]. It follows the same experimental collection protocol and uses the same number of electrodes to obtain sEMG data. The subset DB2 consists of sEMG signals collected from 40 complete subjects performing gesture movements using 12 electrodes. Record each gesture at a sampling rate of 2000 HzHz, repeat each gesture 6 times, and the relaxed state between each repetition is the resting posture. This subset of data contains 49 gestures, including 17 wrist movements and hand postures (Exercise B), 23 grasping and functional movements (Exercise C), and nine9 force patterns (Exercise D). The subset DB4 consists of sEMG signals collected from 10 complete subjects performing gesture movements using 12 electrodes. It contains 52 gestures: Exercise A includes 12 finger movements, Exercise B includes 17 wrist movements and hand postures, and Exercise C includes 23 grasping and functional movements. The detailed information of the two subset includes the number of electrode channels, number of subjects, number of gestures, number of conversations, sampling rate, etc. The number of sessions refers to the number of independent sessions or data collection times recorded during the collection and analysis of sEMG signals.

Data preprocessing

The sEMG is a weak time series signal that primarily reflects the random or non-random bioelectric activity of the neuromuscular system. During the process of collecting sEMG signals, various types of noise interference are also present, mainly including high frequency noise, power frequency interference, and baseline drift. Preprocessing sEMG signals, including denoising, data segmentation, and format conversion, is crucial for subsequent recognition.

Denoising

The NinaPro database is the sEMG signal collected by electrode sensors, and the presence of noise can have a negative impact on the accuracy of feature extraction and gesture recognition [34]. The sEMG is a relatively weak electrical signal, with a peak amplitude of 0–-6 mV obtained from surface electrodes. The useful energy of the signal is in the frequency range of 0–-500 HzHz [35]. Therefore, a fourth-order Butterworth filter with a bandpass boundary of 20–-500 HzHz is used to eliminate low-frequency and high-frequency noise. At the same time, a 50/60 HzHz notch filter is required to eliminate power frequency interference.

Data segmentation and format conversion

Due to the fact that the recorded sEMG signal includes resting posture and action posture, it is usually necessary to segment the activity segment signal [35]. The two common segmentation methods include rough segmentation based on signal envelope and segmentation adjustment based on integrated electromyography (IEMG), the latter providing more accurate signal segmentation positions. This study follows the segmentation method previously studied [24, 44, 47–50] and utilizes a sliding window for signal segmentation. The sliding windows can slide overlapping or non-overlapping, and using overlapping sliding windows can better capture the dynamic changes of signals. By investigating the impact of window length and step size on accuracy, this study selected a fixed window length of 150 ms and a step size of 75 ms, and sets the response time to 300 ms to avoid time delay in real-time systems [23, 51].

Considering that CNN has better recognition performance for two-dimensional images, this study normalizes the signal values to the range of 0– to 1 through grayscale mapping, and then multiplies them by 255 to obtain the corresponding grayscale values. Convert the one-dimensional signal obtained from sliding segmentation into a $T\times 12$ two-dimensional sEMG image, where T is the time frame rate. According to the window length setting, the size of the sEMG image is 300×12. Therefore, sEMG signal gesture recognition can be regarded as an image recognition problem. Figure 1 shows the three steps of data segmentation and format conversion.

Methods

The SE DenseNet architecture proposed in this article for gesture recognition based on sEMG signals is shown in Figure 2. This architecture preprocesses the raw EMG signals and converts them into grayscale images for input. Two dimensional convolution is suitable for processing two-dimensional data such as images and can effectively capture spatial features. The grayscale image is dimension enhanced and sent to a two-dimensional convolution block, so the

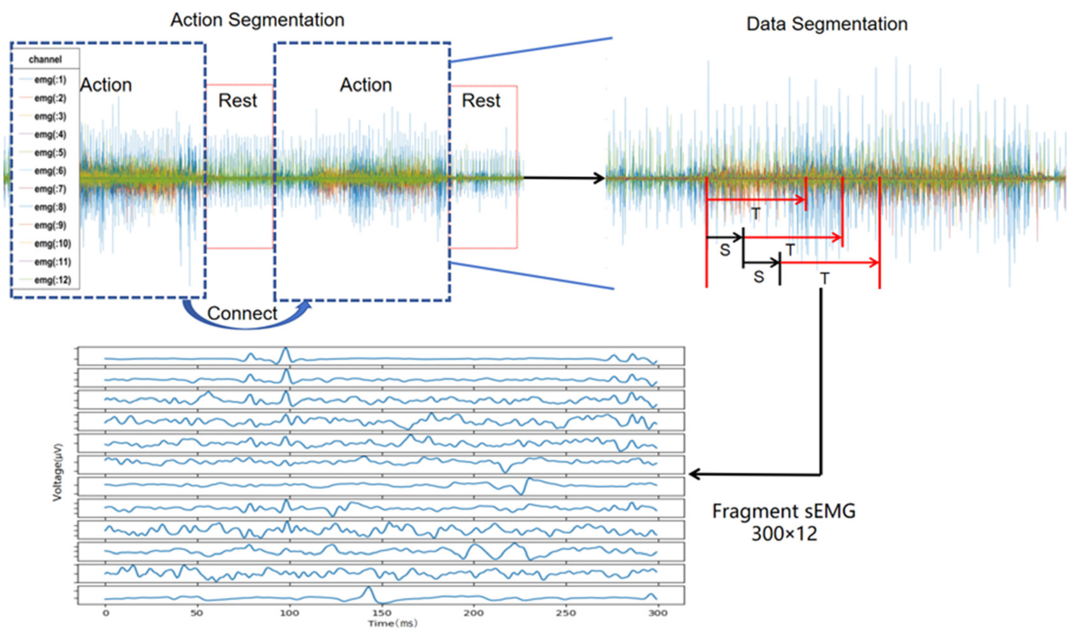

Figure 1: Steps for data segmentation and format transformation.

format of the input EMG image is $H \in R^{T \times E \times C}$, where C represents the number of feature channels. Each two-dimensional convolution block contains a convolution layer, a Rectified Linear Unit (ReLU) activation layer, and a Batch Normalization layer. The SE-DenseNet model consists of two main parts: DenseNet and attention mechanism, which will be discussed in detail later.

DenseNet

DenseNet is an improved CNN based network that computes dense multi-scale features from the convolutional layers of object classifiers. The dense computation of features from a single complete image can significantly accelerate training speed [36]. The DenseNet architecture employs dense connections, linking the output of each layer to the input of all subsequent layers, thereby reducing the number of parameters and computational costs without affecting performance. The densely connected structure enhances the flow of information and gradients, alleviating problems related to gradient vanishing. It also promotes feature reuse, thereby improving the efficiency and accuracy of the network.

The core idea of DenseNet is to achieve more effective feature reuse and gradient propagation by minimizing information loss, thus improving the performance of the network. The formula (1) used to represent the input of the first layer of DenseNet.

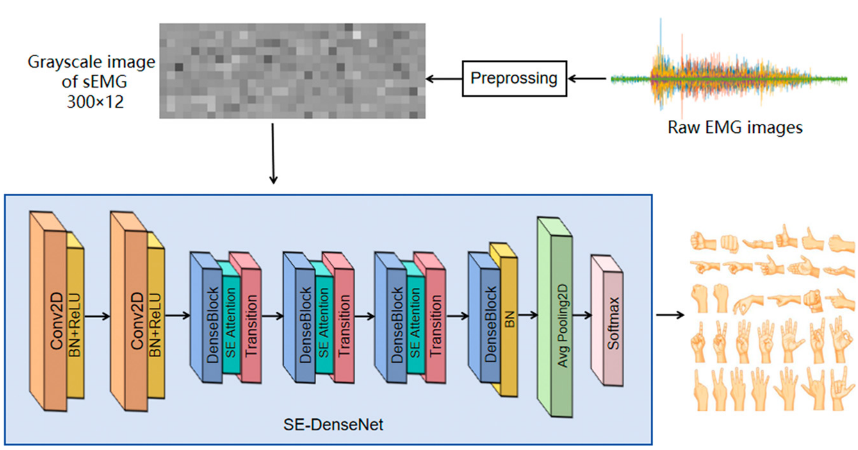

Figure 2: Schematic diagram of sEMG gesture recognition based on SE-DenseNet.

$$x_l = H_l([x_0, x_1, x_2, \ldots, x_{l-1}]) \quad (1)$$

H_l represents a nonlinear transformation function, which includes Batch Normalization (BN) layer, ReLU, and convolutional layer. $[x_0, x_1, x_2, \ldots, x_{l\text{-}1}]$ represents merging outputs from layers 0 to layer l-1.

The DenseNet architecture mainly consists of DenseBlock and Transition components, which utilize dense connections and fewer parameters to reduce computational complexity. The transition module consists of BN layer, ReLU, Convolution layer, and Pooling layer. This Transition module links two adjacent DenseBlocks and reduces the size of the feature map through the pooling layer, highlighting the importance of high-level feature representations in enhancing compression efficiency. This article utilizes a DenseNet network that consists of four DenseBlock modules and three Transition modules.

Attention mechanism

Attention mechanism is a data processing method in machine learning that has been widely applied in various fields of deep learning in recent years, including natural language processing, image recognition, and speech recognition [33, 37]. In 2014, Volodymyr Mnih proposed integrating attention mechanisms into RNN models for image classification, resulting in impressive performance improvements [38]. Attention mechanisms can be divided into channel attention, spatial attention, and mixed domain attention. In the study of channel attention, new architectures continue to emerge. Jie et al. [39] focused on channel relationships in convolutional neural networks and proposed a new architectural unit called "Squeeze and Excitation" (SE) blocks, which are stacked together to form a SENet architecture that adaptively recalibrates the feature responses of channels by simulating their interdependence. Wang et al. [40] designed a non-dimensionality-reducing local cross-channel interaction strategy (Efficient Channel Attention) and an adaptive method for selecting the size of one-dimensional convolution kernels, thereby achieving performance improvement. Qin et al. [41] introduced a multi-spectral channel attention framework called FcaNet, which extends Global Average Pooling (GAP) to enrich feature diversity in the frequency domain.

This work uses SENet to learn global feature information and selectively enhance important features. Firstly, Input X is transformed into feature U through the transformation function F_{tr}, where $X \in R^{h\times w\times c_1}$, $U \in R^{h\times w\times c_2}$. Secondly, the squeezing module F_{sq} uses a global average pool to compress the feature U into $R^{1\times 1\times c_2}$, representing the global distribution of responses on the feature channels. By compressing global spatial information into channel descriptors, the channel dependency problem is alleviated. Then, $F_{ex}(\bullet,w)$ generates weights for each feature channel using parameter w. By reweighting, the weight of the excitation output is considered as the importance of each feature channel. Finally, the weights are applied to the previous feature channels through multiplication to recalibrate the original features.

SE-DenseNet architecture

This work proposes a new network model called SE-DenseNet for gesture recognition of sEMG signals, which mainly consists of four DenseBlocks, three Transitions, and three SENets, as shown in Figure 2.

The EMG image format for the input model is $H \in R^{T\times E\times C}$. Firstly, perform 2D convolution on the EMG image, which facilitates local feature extraction and includes two Conv2D layers: filter=24, kernel size=3, padding='same", stride=1. The main parameters settings for the second Conv2D-layer are: filter=64, kernel size=1, padding='same", stride=1. To accelerate convergence and avoid vanishing gradient problem, this study performs both batch normalization and activation function processing after each two-dimensional convolution. The DenseBlock module in the DenseNet network introduces a hyperparameter called growth rate, which is set to 12. This parameter controls the number of channels added in each convolutional layer, enabling the network to achieve a balance between model complexity and performance. Each DenseBlock is composed of multiple Bottleneck layers. Each Bottleneck is composed of BN layer, ReLU, 1x1convolutional layer, BN layer, ReLU, and 3x3 convolutional layer in sequence. The compression factor of 0.2 is applied in the Transition Layer between DenseBlocks. In this case, the function of 1x1 convolution is to set the number of output channels and achieve dimensionality reduction. Next, this study used a DenseNet network consisting of four DenseBlock modules and three Transition modules, which is defined by seven consecutive operations: BN layer, ReLU, 1x1 convolutional layer, BN layer, ReLU, 3x3 convolutional layer, and Dropout layer. The Dropout layer aims to avoid overfitting issues. Insert SENet between DenseBlock and Transition to learns the importance of each channel and enhance useful functionality. The proposed detailed framework is shown in Table 1.

Table 1: Detailed framework of SE-DenseNet.

Layers	Output size	Connected to
Input	(None, 300, 12, 1)	Conv2D1
Conv2D1	(None, 300, 12, 24)	Conv2D2
Conv2D2	(None, 300, 12, 64)	DenseBlock1
DenseBlock1	(None, 300, 12, 208)	SE block
SE block	(None, 300, 12, 208)	Transition
Transition	(None, 150, 6, 104)	DenseBlock2
DenseBlock2	(None, 150, 6, 248)	SE block
SE block	(None, 150, 6, 248)	Transition
Transition	(None, 75, 3, 124)	DenseBlock2
DenseBlock2	(None, 75, 3, 268)	SE block
SE block	(None, 75, 3, 268)	Transition
Transition	(None, 38, 1, 134)	DenseBlock4
DenseBlock4	(None, 38, 1, 278)	Global AvgPooling
Global AvgPooling	(None, 278)	Dense
Dense	(None, 49)	Output

Table 2: Analysis of classification accuracy (%) for different architectures.

Dataset	Method	Accuracy	Appraisal procedure
NinaPro DB2	Wei et al. [52]	83.7 %	In-session: 1, 3, 4, 6 training
	Zhen et al. [24]	78.86 %	In-session: 1, 3, 4, 6 training
	Zhai et al. [47]	78.71 %	In-session: He first set served as the initial training set
	Zhang et al. [45]	83.29 %	Five-fold cross-validation
	Hu et al. [53]	84.84 %	In-session: 1, 3, 4, 6 training
	Ours	85.93 %	In-session: 1, 3, 4, 6 training
NinaPro DB4	Xu et al. [51]	77.61 %	In-session: 1, 2, 4, 6 training
	Zhang et al. [45]	73.32 %	Five-fold cross-validation
	Nguyen et al. [54]	78.7 %	In-session: 5 training sessions
	Ours	82.39 %	In-session: 1, 3, 4, 6 training

Results and discussion

Training environment and evaluation indicators

To compare the effectiveness of the proposed model, this study adopted the commonly used trial protocol of using the NinaPro database for research [21, 23]. In this intra-session subject evaluation, a deep learning model was trained using partial data from one subject (repetitions 1, 3, 4, and 6) and tested on the remaining data from the same subject (repetitions 2 and 5) [24, 45, 46]. This work uses accuracy, recall, and F1 score as evaluation indicators, and finally averages the accuracy achieved by all subjects to obtain gesture recognition accuracy.

All experiments were implemented on TensorFlow GPU 2.2.0, Python 3.8, and Keras 2.3.1 in Win, and trained on RTX2080ti using cross entropy function as the loss function and Adam optimization algorithm [43, 46, 50]. Set the batch size epoch to 50 and the learning rate to 0.001.

Comparison with baseline model

This study compared the performance of SE-DenseNet model with MV-CNN, DVMSCNN, CNN-RNN and other models. The comparative experimental results are presented in Table 2, and the performance data represents the average value of the subjects.

From Table 2, the proposed model has an accuracy of 85.93 % in classifying all gestures in the NinaPro DB2 dataset, which is 2.64 % higher than the best performance demonstrated by Zhang et al. [45] and 2.23 % higher than Wei et al. [52]. The classification accuracy of all gestures in the NinaPro DB4 dataset is 82.39 %, which is 9.07 % higher than the best performance demonstrated by Zhang et al. [45] and 4.78 % higher than Xu et al. [51].

The model proposed in this article fully utilizes shallow features and combines them with deep features to form richer feature representations, thereby enhancing the sensitivity of the model to detailed information and promoting feature reuse. In addition, introducing the SENet module into the DenseNet architecture allows for feature weighting, enabling the model to more effectively focus on the most important features in the task at hand. Therefore, the model proposed in this study has advantages in feature extraction and information processing.

Figure 3 (a) shows the normalized confusion matrix for 49 gesture recognitions on the NinaProDB2 dataset. The study suggests that the proposed algorithm may mistake the 11th gesture for the 9th gesture, and the confusion is more pronounced in exercise C compared to exercises B and D. Figure 3 (b) shows the confusion matrix between grasping and functional actions (i.e. Exercise C actions), it can be seen that the fourth gesture is often detected as the eleventh gesture with a probability of 41 %, and the eleventh gesture is easily missed as the fourth gesture and the seventeenth gesture. The probability of missing the fourth gesture is the highest, at 41 %.

Discussion

The impact of sliding windows

The size of the sliding window affects time dynamics and computational efficiency. Although smaller windows can

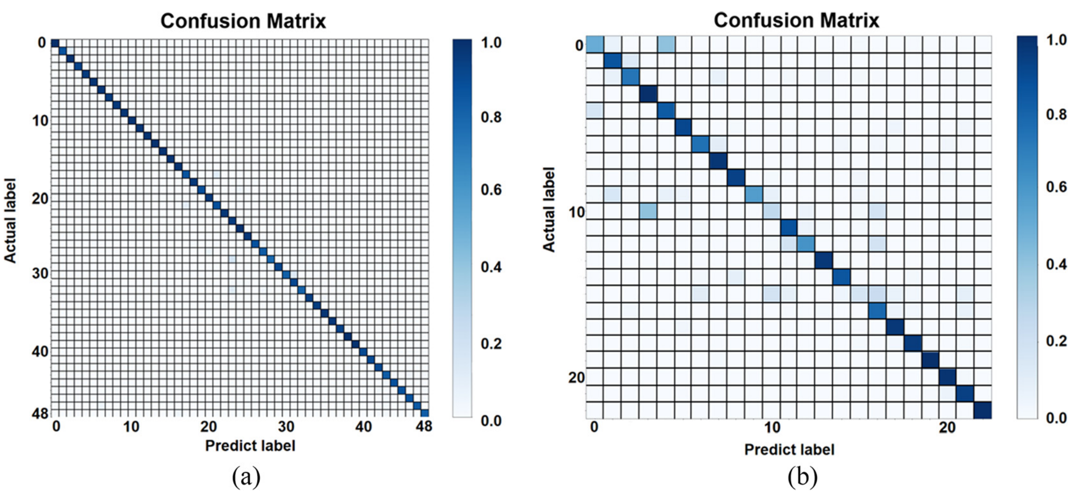

Figure 3: Gesture recognition from sEMG based on the SE-DenseNet network. (a) Normalized confusion matrix for all gesture recognition in NinaPro DB2. (b) The confusion matrix between grasping and functional actions (i.e. Exercise C actions).

capture dynamic changes faster, they may lead to information loss and reduced data processing efficiency. On the contrary, larger windows capture signal features over a longer period of time, but may respond slower to dynamic changes. Based on previous studies [23, 24, 26, 42–44], this study compared the effects of different window sizes such as 100 m sms, 150 m sms, and 200 m sms on recognition accuracy while maintaining a 50 % overlap rate to partially compensate for information loss caused by window size.

The height of each bar in Figure 4 represents the average accuracy of the corresponding accuracy. As the window length increases, there is no significant change in recognition rate. Considering the balance between recognition performance and recognition time, this study chose a sliding window size of 150 m sms in the experiment.

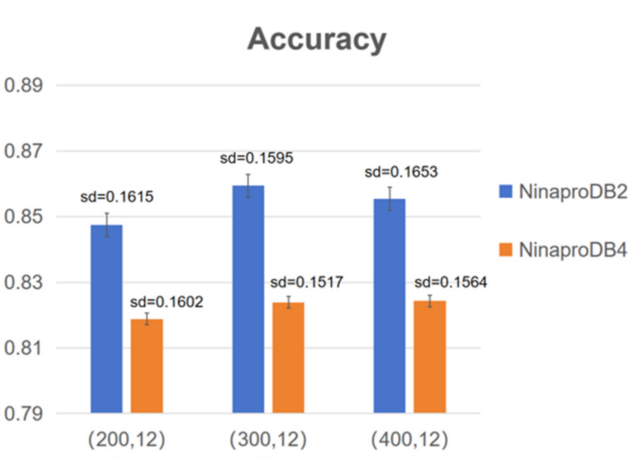

Figure 4: The impact of sliding window size on SE-DenseNet performance.

The impact of different attention mechanism

Attention mechanisms are mainly divided into channel attention and spatial attention, and different attention mechanisms can affect the performance of the model. This study compared the impact of channel attention mechanism ECA and mixed attention mechanism CBAM on model performance. Table 3 shows the performance of the two attention mechanisms based on the SE-DenseNet model in terms of average accuracy, and the hybrid model of SE-DenseNet performs better. This is because SE-DenseNet can more effectively capture key features when processing complex data, and this feature enhancement ability makes it perform better in feature extraction and classification tasks. CBAM first calculates channel attention, and then calculates spatial attention. Although it can capture the global dependencies of feature maps, its design is relatively complex and increases computational overhead. ECA aims to achieve channel attention with fewer parameters and computational complexity, but its ability to express features may not be as comprehensive as SE-DenseNet, and it cannot fully capture the global dependencies between channels.

Table 3: The performance of different attention mechanisms.

Dataset	Model		
	SE-DenseNet	ECA-DenseNet	CBAM-DenseNet
NinaPro DB2	85.93 %	83.27 %	82.05 %
NinaPro DB4	82.39 %	81.33 %	80.56 %

Table 4: Performance comparison of different numbers of dense connection blocks.

Dataset	Basic model		
	DenseNet-101	DenseNet-121	DenseNet-169
NinaPro DB2	85.93 %	85.17 %	85.21 %
NinaPro DB4	82.39 %	82.41 %	82.42 %

The impact of the number of dense blocks in DenseNet

The performance of DenseNet is affected by the number of dense blocks and transition block layers. This study compared DenseNet-101, DenseNet-121, and DenseNet-169 as the baseline models. The difference between the network structures of the three basic models lies in the number of dense connection blocks. According to the final results in Table 4, it can be seen that DenseNet-101 performs better. Based on the experimental results, DenseNet-101 was ultimately chosen as the base model for this study.

Conclusions

This article proposes an SE-DenseNet network structure for recognizing sEMG signals. DenseNet utilizes dense connection blocks to promote feature reuse, which helps preserve shallow features, ensures efficient parameter utilization, enhances model learning ability, and alleviate the problem of gradient vanishing. Based on the comparison of ablation experiments, DenseNet-101 was chosen as the basic model in this study. In order to improve recognition rate, this study inserts SENet between DenseBlock and Transition layers, and uses attention mechanism to capture spatial relationships between features and establish mutual connections between channels. This work validates the SE-DenseNet model on the NinaPro DB2 and DB4 datasets, and the experimental results show that the proposed algorithm performs well in sEMG signal gesture recognition. Specifically, the final recognition rate of all gestures was 85.93 % for NinaPro DB2, and 82.39 % for NinaPro DB4.

However, this study also has certain limitations and deserves further discussion. Although the SE-DenseNet model shows promising results, it may encounter difficulties with complex gestures or gestures with similar properties, which may lead to classification errors. In addition, the performance of the model may be affected by signal quality, noise, and individual differences in sEMG signals. Future research could explore hybrid architectures that integrate SE-DenseNet with other advanced models such as transformer networks to improve the accuracy and robustness of gesture recognition. In addition, adopting technologies such as data augmentation, transfer learning, and ensemble methods can bring further improvements. Overall, addressing these limitations and exploring potential avenues for future research could make a significant contribution to the development of more effective and reliable gesture recognition systems based on sEMG signals.

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Conflict of interest: The authors state no conflict of interest.
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Data availability: The datasets presented in this study can be found in the online repository, and the name of the repository can be found below: https://ninapro.hevs.ch/.

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